

The content extracted is as follows:

2. Preliminaries

end if end for end while τ_{max} stops the tree extraction process when one of two criteria is met: no more nodes need to be further expanded because their entropy is low enough (they contain almost exclusively instances of a single class) or a predefined limit of the tree size (the number of nodes) is reached. While τ_{repan} was designed to explain neural networks as the oracle, it is model-agnostic and can be used to explain any other classification model.

In order to evaluate the performance of algorithms like τ_{repan} that approximate an initial machine learning model with a so-called surrogate, we measure the accuracy and the fidelity of the resulting surrogate, i.e. the decision trees.

3. TREPANReloaded

We use this information content measure to prioritise features that are more general (thus presenting less information content), as our hypothesis is that the decision tree becomes more understandable when it uses more general concepts.

5. Experimental evaluation

In the fourth task, the empowerment task, respondents were shown a particular statement and had to decide what action (if any) they could take to change the decision of the model (e.g., You are a male and you have 3 children.

6. Discussion

At this point, one might wonder why we should bother to create surrogate decision trees from black-box models, rather than inferring them directly from data.

As we discussed above, in this approach the accuracy of the resulting trees (i.e., their ability to match ground truth) is less relevant than their fidelity (i.e., their ability to match the behaviour of the model we intend to explain).

These concepts could then be used to create an interpretable model explaining what the CNNs learned. The method proposed in this paper and other approaches (e.g., LIME [55], SHAP [60,40]) explain the role of features in decision making by creating surrogate models that are interpretable.

7. Related work

Whilst local explanation methods aim to explain specific decisions, global approaches aim to provide a more comprehensive understanding of how a model works.

LIME perturbs the data for a given decisions and, after feeding the black-box model with perturbed data, creates a new linear model from the predictions.

This achieves a better fidelity with respect to the black-box models decision boundaries.

Several local explanation methods have been proposed based on Shapley values (e.g., [40,60]), which quantify the influence of a feature on a decision by training models on all subsets of features and observing their output depending on the presence of the given feature.

The authors propose ConceptTrees, a version of τ_{repan} that builds on features belonging to concepts in the extraction of a decision tree.

Dhurandhar et al. [20] propose ProfWeight, a method designed to transfer information from a pre-trained deep learning model to a simpler one, e.g., a simpler neural network or a decision tree.

The main motivation behind this approach is not only to create more explainable decision models.

The technique generates an interpretable approximation of the original model, in the form of a decision tree.

8. Conclusion and future works

It is an extension of τ_{repan} , an algorithm that extracts global post-hoc explanations of black-box models in the form of decision trees.